

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
PART TEST – III
PAPER –2
TEST DATE: 19-12-2021

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

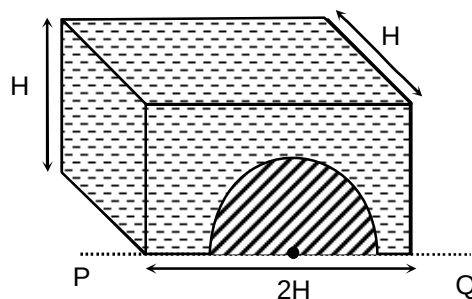
Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. A liquid of density ' ρ ' is filled in a cuboidal container of dimension ($H \times H \times 2H$) as shown in the figure. A semicircular gate is made at the bottom of container of radius $R = \frac{H}{2}$. Let F is the force acting on the gate due to the liquid and τ is the torque acting on the gate due to the liquid about a line PQ. Then choose the correct option(s).



- (A) $F = \frac{H^3 \rho g}{4} \left(\frac{\pi}{4} - \frac{1}{3} \right)$
- (B) $F = \frac{H^3 \rho g}{2} \left(\frac{\pi}{4} - \frac{1}{6} \right)$
- (C) $\tau = \frac{H^4 \rho g}{4} \left(\frac{1}{3} - \frac{\pi}{32} \right)$
- (D) $\tau = \frac{H^3 \rho g}{3} \left(\frac{1}{3} - \frac{\pi}{16} \right)$

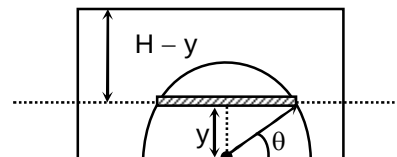
Ans. B, C

Sol. $dF = p dA = \rho g (H - R \sin \theta) \cdot 2R \cos \theta \cdot R \cos \theta d\theta$

$$F = 2R^2 \rho g \int_0^{\pi/2} (H - R \sin \theta) \cos^2 \theta d\theta = 2R^2 \rho g \left(\frac{H\pi}{4} - \frac{R}{3} \right)$$

$$d\tau = dF y$$

$$\tau = 2R^3 \rho g \int_0^{\pi/2} (H - R \sin \theta) \cos^2 \theta \sin \theta d\theta = 2R^3 \rho g \left(\frac{H}{3} - \frac{\pi R}{16} \right)$$



2. Consider a hypothetical hydrogen like atom. The wavelength of spectral line from p^{th} orbit to ground state is given by $\lambda = \frac{3500 p^2}{p^2 - 1}$ in Å. Choose the correct option(s) for considered hypothetical atom. ($hc = 1240 \text{ eV-nm}$)
- (A) The ionization potential of considered hypothetical atom is 3.54 V approximately.
- (B) The ionization potential of considered hypothetical atom is 0.354 V approximately.
- (C) The wavelength of least energetic photon in the spectral line series is 466.67 nm approximately.

- (D) The wavelength of maximum energetic photon in the spectral line series is 350.00 nm approximately.

Ans. A, C, D

Sol. $\lambda = \frac{3500}{1 - \frac{1}{p^2}}$

$$\lambda_{\max} = \frac{3500}{1 - \frac{1}{4}} = 466.67 \text{ nm}$$

3. A Searle's experiment, a wire of diameter $d = 0.060 \text{ cm}$ is used to find its young's modulus. The length of given wire is $L = 200 \text{ cm}$ and when weight of mass $m = 30.0 \text{ kg}$ is used in experiment, extension in the length of wire was found to be 0.200 cm . Then choose the correct option(s).

- (A) Maximum permissible error in young's modulus of wire is 4.66%.
- (B) Maximum permissible error in young's modulus of wire is 3%.
- (C) The least count of the instrument by which length of the wire is to be measured is 1 cm.
- (D) The least count of the instrument by which diameter of the wire is to be measured is 0.001 cm.

Ans. A, C, D

Sol. $\left| \frac{\Delta Y}{Y} \right|_{\max} = \frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{2\Delta d}{d} + \frac{\Delta x}{x}$

$$= \frac{0.1}{30} + \frac{1}{200} + \frac{2(0.001)}{0.060} + \frac{0.001}{0.200} = 0.0466$$

$$\left| \frac{\Delta Y}{Y} \right|_{\max} = 4.66\%$$

4. A cylindrical string is used in guitar. The string is to be made of 11.0 gm and density of string is $5 \times 10^3 \text{ kg/m}^3$. The string can withstand a breaking stress $14 \times 10^6 \text{ N/m}^2$. String can be used under a tension of 44 newton without breaking. A stationary wave is formed on the guitar string. Now choose the correct option(s).

- (A) Length and radius of the longest and thinnest string are 70 cm and 1 mm respectively.
- (B) Length and radius of the shortest and thickest string are 70 cm and 1 mm respectively.
- (C) When the standing wave is formed in the guitar string, then the particle of the string at the maximum displacement from mean position have maximum potential energy.
- (D) The frequency of the fourth overtone formed in the guitar string is 189 Hz approximately.

Ans. A, D

Sol. Breaking stress = $\frac{F}{A} = \frac{F}{\pi r_{\min}^2}$

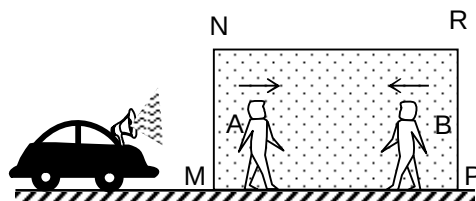
$$r_{\min} = \sqrt{\frac{44}{\pi \times 14 \times 10^6}} = 10^{-3} = 1 \text{ mm}$$

$$m = \rho(\pi r_{\min}^2 L_{\max})$$

$$L_{\max} = \frac{m}{\pi \rho r_{\min}^2} = \frac{11 \times 10^{-3}}{\pi \times 5 \times 10^3 \times (10^{-3})^2} = 0.7 \text{ m}$$

$$f = \frac{5v}{2L} = \frac{5}{2L} \sqrt{\frac{F}{\mu}} = 189 \text{ Hz}$$

5. A car is moving with a velocity $\frac{3v}{10}$ towards a medium MNRP as shown, where 'v' is the speed of sound in air. Two men A and B are moving towards each other in the medium with the velocities $\frac{v}{6}$ and $\frac{v}{9}$ respectively. The speed of sound in medium is $7v$ and medium is kept at rest. The car is emitting a sound of frequency f . Then choose the correct option(s).



- (A) The wavelength received by the man A is $\frac{49v}{10f}$.
- (B) The wavelength received by the man A is different from the wavelength received by man B.
- (C) The ratio of frequency observed by the man A to that observed by the man B is $\frac{123}{128}$.
- (D) The frequency observed by the man B is $\left(\frac{640}{441}\right)f$.

Ans. A, C, D

Sol. Frequency of surface MN (f') = $\frac{10f}{7}$

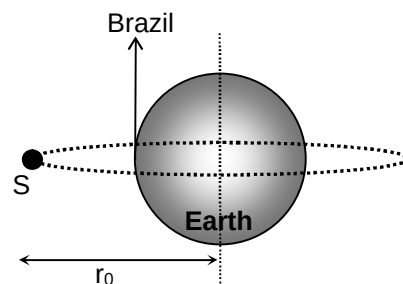
$$\lambda_A = \frac{7v}{f'} = \frac{49v}{10f}$$

$$\lambda_B = \frac{7v}{f'} = \frac{49v}{10f}$$

$$f_A = \left(\frac{7v - \frac{v}{6}}{7v} \right) f' = \left(\frac{205}{147} \right) f$$

$$f_B = \left(\frac{7v + \frac{v}{9}}{7v} \right) f' = \left(\frac{640}{441} \right) f$$

6. Consider a satellite 'S' which is just above the Brazil and rotating on equatorial plane of earth. The sense of rotation of earth and satellite are similar means both are rotating from west to east. The time period of satellite is 6 hours. At $t = 0$, satellite is just above the Brazil. Now choose the correct option(s).



- (A) The time after which the satellite will be above the Brazil again is 8 hours.
- (B) The above satellite may be a geostationary satellite.
- (C) The height of geostationary satellite above the sea level is 35,800 km approximately.
- (D) If the distance from the centre of earth to the satellite 'S' becomes $4r_0$, then the time after which the satellite will be above the Brazil again is 24 hours.

Ans. A, C

Sol. Angular velocity of earth $\omega_1 = \frac{\pi}{12}$ rad/hr

Angular velocity of satellite $\omega_2 = \frac{\pi}{3}$ rad/hr

$$(\omega_2 - \omega_1)t = 2\pi$$

$$t = 8 \text{ hours}$$

→ for option 'B'

$$T^2 \propto r^3$$

$$T' = 48 \text{ hours}$$

$$\omega' = \frac{\pi}{24} \text{ hours}$$

$$(\omega' - \omega_1)t = 2\pi$$

$$t = 48 \text{ hours}$$

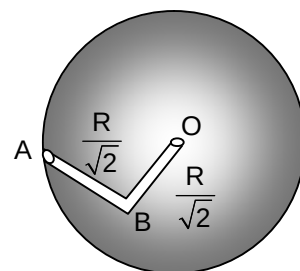
Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

An isosceles right angle tunnel ABO is made inside the earth from surface of earth to its centre as shown in the figure. The distance between the points A and B and that between the points B and O is $\frac{R}{\sqrt{2}}$, where R is the

radius of earth. The surface of tunnel is assumed to be frictionless. A small particle of mass m_0 is released from point A. The collision between the particle and the tunnel is assumed to be perfectly inelastic. Take mass of earth is M, Radius of earth $R = 6400 \text{ km}$ and $g = 10 \text{ m/s}^2$. Now answer the following questions.



7. The time taken by the particle to move in the tunnel from A to O.

- (A) $1600 \pi \text{ sec}$
- (B) $800 \pi \text{ sec}$
- (C) $400 \pi \text{ sec}$
- (D) $200 \pi \text{ sec}$

Ans. B

8. Work done by gravitational field of earth on the particle to reach from B to O.

- (A) $\frac{Gm_0M}{2R}$
- (B) $\frac{Gm_0M}{8R}$
- (C) $\frac{Gm_0M}{4R}$
- (D) $\frac{Gm_0M}{6R}$

Ans. C

Sol. (Q.7-8)

$$t_{AB} = \frac{T}{4}$$

$$t_{BO} = \frac{T}{4}, \text{ where } T = 2\pi\sqrt{\frac{R}{g}}$$

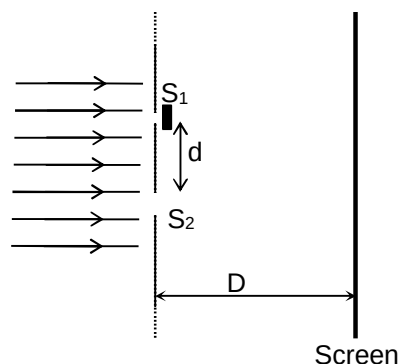
$$t = t_{AB} + t_{BO} = \frac{T}{2} = \pi\sqrt{\frac{R}{g}} = 800\pi \text{ sec}$$

$$W_{gr} = -m_0(V_0 - V_B)$$

$$= -m \left(-\frac{3}{2} \frac{GM}{R} + \frac{5}{4} \frac{GM}{R} \right) = \frac{Gm_0M}{4R}$$

Paragraph for Question Nos. 09 and 10

A monochromatic light of wave length $\lambda = 6000 \text{ \AA}$ is used and incident on two slits S_1 and S_2 as shown. The width of slits S_1 and S_2 are different in such a way the intensity on the screen due to the slit S_2 is 9 times the intensity due to the slit S_1 on the screen. The distance between the two slits S_1S_2 is $d = 1\text{mm}$ and the distance between the slits and the screen is $D = 2\text{m}$. A mica sheet of refractive index $\left(\mu = \frac{4}{3}\right)$ and thickness $t = 0.1 \text{ }\mu\text{m}$ is kept in front



of slit S_1 . Assuming both the slits behave like coherent sources. The minimum distance between the two points on the screen which have intensities equal to the $\left(\frac{1}{4}\right)^{\text{th}}$ of the maximum intensity on the screen is 'a' mm. The minimum distance between the two points on the screen which have intensities equal to $\left(\frac{13}{16}\right)$ times the maximum intensity on the screen is 'b' mm.

9. The value of 'a' is

Ans. 1.20

10. The value of 'b' is

Ans. 0.80

Sol. (Q.9-10)

$$I = I_0 + 9I_0 + 2\sqrt{I_0 9I_0} \cos \phi$$

$$I = 10I_0 + 6I_0 \cos \phi \quad \dots(i)$$

$$I_{\max} = 16I_0, I_{\min} = 4I_0$$

Sol. (9). $\frac{16I_0}{4} = 10I_0 + 6I_0 \cos \phi$

$$\cos \phi = -1$$

$$\phi = \pi, 3\pi, 5\pi \dots\dots\dots$$

$$\frac{2\pi y}{\beta} = \pi, 3\pi, 5\pi \dots\dots\dots$$

$$\text{So, } y_2 - y_1 = \beta = \frac{D\lambda}{d} = 1.2 \text{ mm}$$

$$a = 1.2 \text{ mm}$$

Sol.(10). $(16I_0) \frac{13}{16} = 10I_0 + 6I_0 \cos \phi$

$$\cos \phi = \frac{1}{2}$$

$$\phi = \frac{\pi}{3}, \frac{5\pi}{3}, \dots\dots\dots$$

$$\frac{2\pi y}{\beta} = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$$

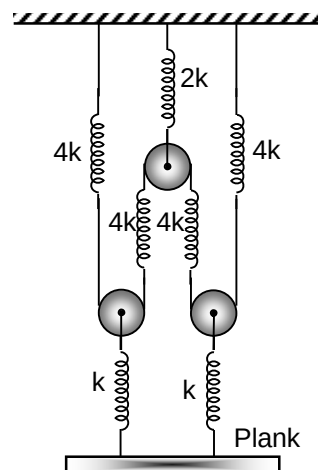
$$y_2 - y_1 = \frac{2\beta}{3} = \frac{2}{3} \frac{D\lambda}{d} = 0.8 \text{ mm}$$

$$b = 0.8 \text{ mm}$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. Seven ideal massless springs and three smooth massless pulleys are connected with a plank of mass M in vertical plane as shown in the figure. Plank is slightly displaced from equilibrium position in vertical plane such that plank always remain in horizontal position. The spring constant of springs are shown in the figure. If angular frequency of oscillation of plank is $\sqrt{\frac{8\lambda k}{11m}}$, then find the value of λ .



Ans. 2

Sol.
$$x = \frac{2x_3 + x_2}{4} + x_1$$

$$2kx_3 = 2kx_2$$

$$2kx_2 = kx_1$$

On solving $x_1 = \frac{8}{11}x$

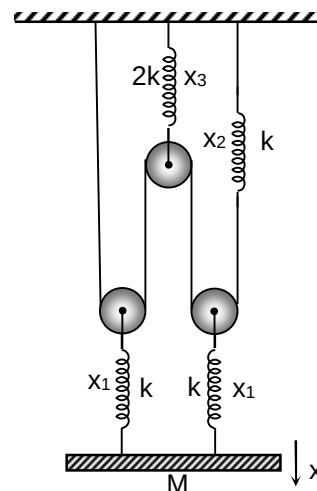
$$F_r = 2kx_1 = \frac{16}{11}kx_1$$

$$\omega = \sqrt{\frac{16k}{11m}}$$

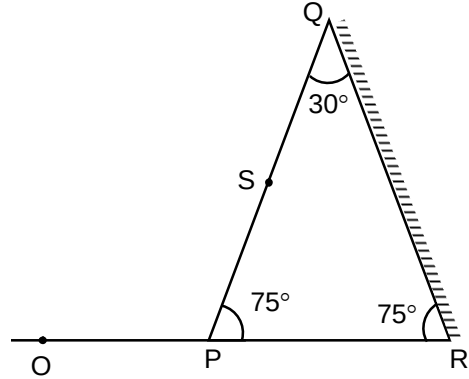
...(i)

...(ii)

...(iii)



12. PQR is an isosceles prism in which point 'S' is the mid point of the face PQ. The refractive index of the prism is $\sqrt{3}$ and its face QR is silvered. 'O' is a point source. A light ray has been sent from the source 'O' to point 'S' in such a way that it comes back to point 'O'. It is given that $PR = \frac{20}{\sqrt{3}+1}$ cm. Find the distance (in cm) of the source 'O' from the point 'P'.



Ans. 5

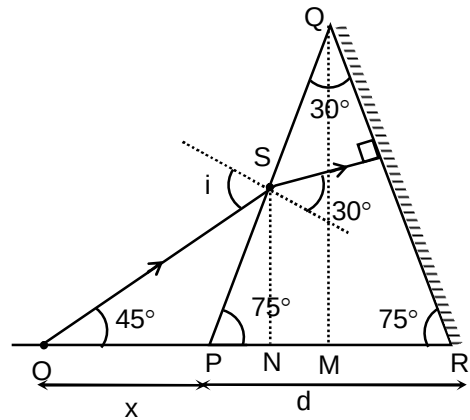
Sol. $1 \times \sin i = \sqrt{3} \times \sin 30^\circ$
 $i = 60^\circ$

$$\tan 75^\circ = \frac{SN}{d/4} \quad \dots(i)$$

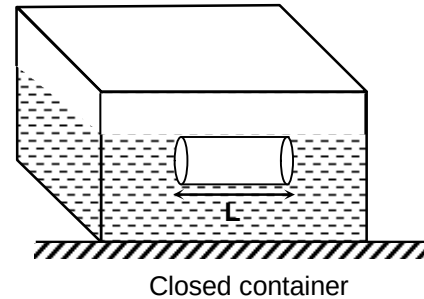
$$\tan 45^\circ = \frac{SN}{x + \frac{d}{4}} \quad \dots(ii)$$

From (i) and (ii)

$$x = \frac{d}{4}(\sqrt{3} + 1)$$



13. A liquid of density $\rho = 5 \times 10^3 \text{ kg/m}^3$ is filled in a closed container. A cylinder of length $L = 2$ meter and radius $r = 1/2$ meter is fully submerged and floats on the surface of liquid as shown in the figure. Take minimum pressure in the container is zero and acceleration due to gravity $g = 10 \text{ m/s}^2$. The force due to pressure of liquid on the lower half surface of the cylinder is $\frac{25}{7\lambda} \times 10^5 \text{ N}$. Find the value of ' λ '. (Take $\pi = \frac{22}{7}$)



Ans. 4

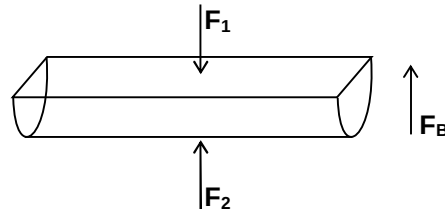
Sol. Draw F.B.D. of half cylinder

$$F_2 - F_1 = F_B$$

$$F_2 = F_B + F_1$$

$$= \frac{\pi r^2}{2} \cdot L \rho g + (\rho g r) 2rL = \rho g L r^2 \left[\frac{\pi}{2} + 2 \right]$$

$$= \frac{25}{28} \times 10^5 \text{ N/m}^2$$



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15
Question Stem

A screw gauge has 100 divisions on the circular scale and linear distance travelled by the head scale is 5 mm in 10 complete rotation of circular scale. When nothing is placed between its studs, the zero of circular scale is 26 divisions above the reference line, it means zero error is –26 divisions on the circular scale. Now if the diameter of a wire is measured by keeping the wire between its studs, the main scale reading is 6 divisions and 48th division of circular scale coincides with the reference line. If the negative zero error of screw gauge is x mm and the measured diameter of the wire is y mm.

14. The value of x is

Ans. 0.13

15. The value of y is

Ans. 3.37

Sol. (Q.14-15)

$$\text{Pitch} = \frac{5 \text{ mm}}{10} = 0.5 \text{ mm}$$

$$\text{Least count} = \frac{0.5}{100} = 0.005 \text{ mm}$$

$$\text{Negative zero error} = 26 \times 0.005 = 0.13 \text{ mm}$$

$$\text{Diameter of the wire measured, } d = 6 \times 0.5 \text{ mm} + 48 \times 0.005 + 26 \times 0.005 = 3.37 \text{ mm}$$

Question Stem for Question Nos. 16 and 17
Question Stem

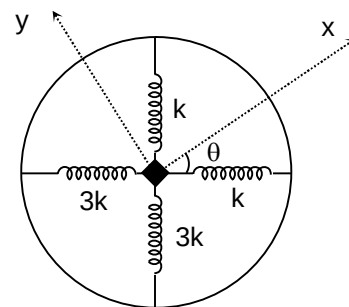
Four ideal massless springs of spring constant k, k 3k and 3k are connected with a ring of mass 3m. The ring has uniform mass density and all the springs are in natural length initially. The other ends of springs are connected with a block of mass m. The system is kept on a frictionless horizontal surface. The block is slightly displaced along the positive x-axis in such a way that the centre of the ring is at origin and co-ordinates of block are (a₀, 0) and then system is released from rest. The motion of the block is simple harmonic motion with a time

period $T = \lambda_1 \pi \sqrt{\frac{\lambda_2 m}{\lambda_3 k}}$. The position vector of the centre of the ring

relative to the origin as a function of time is given by

$$\frac{a_0}{c_1} \left[c_2 - \cos \sqrt{\frac{c_3 k}{c_4 m}} t \right].$$

16. The value of $\left(\frac{\lambda_2 + \lambda_3}{\lambda_1} \right)$ is



Ans. 9.50

17. The value of $\left(\frac{C_3 + C_4}{C_1 C_2}\right)$ is

Ans. 4.75

Sol. (Q.16-17)

Let 'x' is the displacement of block along x-axis, then restoring force will be

$$F_r = kx \cos^2 \theta + 3kx \cos^2 \theta + kx \sin^2 \theta + 3kx \sin^2 \theta \\ = 4kx \cos^2 \theta + 4kx \sin^2 \theta = 4kx \quad \dots(i)$$

$$\text{So, } T = 2\pi \sqrt{\frac{\mu}{k_{eq}}}, \quad \text{where } \mu = \frac{m \cdot 3m}{m + 3m}$$

$$= 2\pi \sqrt{\frac{\frac{3}{4}m}{4k}} = 2\pi \sqrt{\frac{3m}{16k}}$$

$\Rightarrow$ Centre of ring will accelerate about the centre of mass of the system

$$x = \frac{a_0}{4} \left[1 - \cos \sqrt{\frac{16k}{3m}} t \right]$$

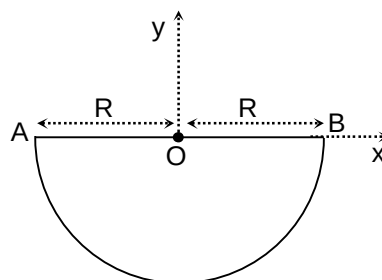
Question Stem for Question Nos. 18 and 19

Question Stem

A wire of uniform linear mass density is bent in the form of semi-circular loop of radius R. Semicircular loop is hinged at origin in a vertical plane as shown in the figure. It can freely rotate about the z-axis in the vertical plane. Now, a small angular displacement of 2° is given to the loop and released from rest such that it oscillate about the z-axis in the vertical xy plane as shown in the figure. During the subsequent motion, the maximum velocity of point B is

found to be $\frac{\pi}{n_1} \sqrt{\frac{n_2 g R}{(n_3 + n_4 \pi)}}$, where n_1, n_2, n_3 and n_4 are +ve

constants. Air resistance is neglected and 'g' is acceleration due to gravity.



18. The value of $\left(\frac{n_1}{n_2 \cdot n_3}\right)$ is

Ans. 7.50

19. The value of $\left(\frac{n_2 n_3 n_4}{n_1}\right)$ is

Ans. 0.40

Sol. (Q.18-19)

$$I_z = \lambda(2R) \frac{(2R)^2}{12} + \lambda(\pi R)R^2 = \frac{\lambda R^3}{3}(2 + 3\pi) \quad \dots(i)$$

$$d = \left(\frac{2R}{2 + \pi} \right) \quad \dots(ii)$$

$$T = 2\pi \sqrt{\frac{I_z}{mgd}} = 2\pi \sqrt{\frac{(2 + 3\pi)R}{6g}} \quad \dots(iii)$$

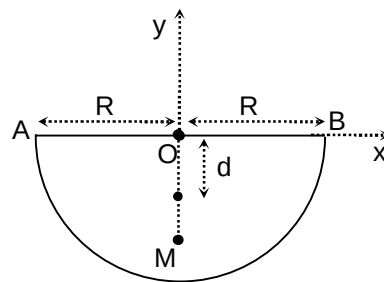
$$\Rightarrow \theta_0 = 2^\circ = \frac{\pi}{90}$$

$$\theta = \theta_0 \cos \omega t$$

$$\text{Angular velocity} = \frac{d\theta}{dt} = -\theta_0 \omega \sin \omega t$$

$$v_B = R \cdot \frac{d\theta}{dt} = -\theta_0 \omega R \sin \omega t$$

$$(v_B)_{\max} = R\theta_0 \omega = \frac{\pi}{90} \sqrt{\frac{6gR}{(2 + 3\pi)}}$$



Chemistry

PART – II

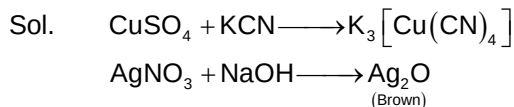
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. Correct statement(s) among the following is/are.

- (A) PH_3 reduces AgNO_3 to Ag
- (B) AgCl is soluble in KCN
- (C) CuSO_4 on reaction with KCN for $\text{K}_2[\text{Cu}(\text{CN})_4]$
- (D) AgNO_3 on reaction with NaOH form brown residue

Ans. A, B, D



21. Which statement(s) is/are correct?

- (A) Higher the oxidation state of metal more stable is the complex
- (B) Chelating ligand form more stable complex as compared to monodentate ligands
- (C) All the octahedral complexes of Ni^{2+} are outer orbital complexes
- (D) $(\text{SCN})^-$ shows linkage isomerism in co-ordination compounds

Ans. A, B, C, D

22. Which of the followings ores are concentrated by froth floatation process?

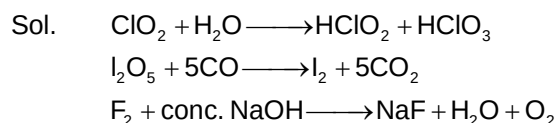
- (A) Copper pyrites
- (B) Zinc blende
- (C) Pyrolusite
- (D) Cinnabar

Ans. A, B, D

Sol. Sulphide ores are concentrated by Froath-Floatation method.
 Pyrolusite is MnO_2 .

23. Which of the followings are correct statements?
- (A) ClO_2 is the anhydride of HClO_2 and HClO_3
- (B) I_2O_5 is used in the quantitative estimation of CO
- (C) Cl_2 on reaction with hot conc. NaOH produces O_2 gas
- (D) F_2 on reaction with hot conc. NaOH produces O_2 gas.

Ans. A, B, D



24. In which of the following species d – d transition do not account for it's colour?
- (A) $\text{Cr}_2\text{O}_7^{2-}$
- (B) CrO_4^{2-}
- (C) CrO_2Cl_2
- (D) MnO_4^-

Ans. A, B, C, D

25. Which statement(s) is/are true?
- (A) Schottky defect is more common in a solid having high radius ratio.
- (B) Frenkel defect is observed in solid having low co-ordination number or low radius ratio.
- (C) In Schottky defect density remain same.
- (D) In Frenkel defect density decreases

Ans. A, B

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

A white compound (A) on heating gives residue and two gas (B) and (C). Gas (B) burns with blue flame. Gas (C) turns lime water milky and residue aqueous solution is alkaline. Compound (A) impart brick-red colour to flame

26. Compound [A] is

- (A) CaCO_3
- (B) CaC_2O_4
- (C) BaC_2O_4
- (D) BaCO_3

Ans. B

27. Gas [B] is

- (A) CO_2
- (B) SO_2
- (C) CO
- (D) SO_3

Ans. C

Paragraph for Question Nos. 28 and 29

Compressibility factor of real gas can be written as

$$Z = \frac{P\bar{V}}{RT} = 1 + \frac{B}{V} + \frac{C}{V^2} + \frac{D}{V^3} \dots\dots\dots$$

This is virial equation and B, C and D are second, third, fourth virial coefficient.

At Boyle's temperature second virial coefficient is zero. $Z > 1$ repulsive force dominates and $Z < 1$ attractive force dominates.

28. Compressibility factor for real gas at high pressure is

- (A) 1
- (B) $1 - \frac{Pb}{RT}$
- (C) $1 + \frac{Pb}{RT}$
- (D) $1 + \frac{RT}{Pb}$

Ans. C

29. Which of the following statement(s) is/are correct?

- (A) When $Z > 1$ real gas are difficult to compress.

- (B) When $Z > 1$ real gas easier to compress.
- (C) Compressibility factor for real gas at critical state is $\frac{3}{8}$.
- (D) Z for real gas at low pressure is $\left(1 - \frac{a}{VRT}\right)$.

Ans. A, C, D

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. The effective atomic number (EAN) of $M(\text{CO})_x$ is 36. The atomic number of M is 26. Find the value of x .

Ans. 5

Sol. $36 = 26 + 2 \times x$
 $x = 5$.

31. 3.55 gram of bleaching powder when treated with acetic acid and excess of KI , liberated iodine required 60 ml 0.05 N $\text{Na}_2\text{S}_2\text{O}_3$ solution. In this reaction available Cl_2 is $x\%$. Find x ?

Ans. 3

Sol. Meq of $\text{Cl}_2 = 60 \times 0.05 = 3$
 $\text{Wt. of } \text{Cl}_2 = \frac{3 \times 35.5}{1000}$
 $\% \text{available } \text{Cl}_2 = \frac{3 \times 35.5}{1000 \times 3.55} \times 100\%$
 $= 3\%$
 $x = 3$

32. The number of atoms in 100 gram of an FCC crystal with density $d = 10 \text{ gram cm}^{-3}$ and edge length as 200 pm is $x \times 10^{24}$. The value of ' x ' is

Ans. 5

Sol. $\rho = \frac{Z M}{N_a a^3}$
 $M = \frac{10 \times 6.02 \times 10^{23} \times (200 \times 10^{-10})^3}{4}$
 $= 12.04$
 Number of atom in 100 gram is $= \frac{6.02 \times 10^{23}}{12.04} \times 100$
 $= 5 \times 10^{24}$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34**Question Stem**

A person inhales 640 gram of O_2 per day. If all the O_2 is used for converting sugar into CO_2 and H_2O . 'x' kg of sugar ($C_{12}H_{22}O_{11}$) is consumed in the body and the heat liberated is $y \times 10^3$ kJ. Heat of combustion of $C_{12}H_{22}O_{11}$ is -5640 kJ/mol.

33. Find x _____

Ans. 0.57

Sol. $C_{12}H_{22}O_{11} + 12O_2 \longrightarrow 12CO_2 + 11H_2O$
 Moles of $C_{12}H_{22}O_{11}$ per day = $\frac{20}{12}$ mole.
 $= \frac{20}{12} \times 342 = 570$ gram
 $= 0.57$ kg
 $x = 0.57$

34. Find y _____

Ans. 9.40

Sol. Heat liberated $\frac{20}{12} \times 5640$
 $= \frac{5}{3} \times 5640$
 $= 9.40 \times 10^3$ kJ
 $y = 9.40$.

Question Stem for Question Nos. 35 and 36**Question Stem**

During electrolysis process of aqueous NaCl, 710 gram of Cl_2 was produced at anode. And 'x' kg of NaOH as well as $y \times 10^2$ litres (S.T.P.) of H_2 gas at cathode obtained during this process.

35. Find the value of x _____

Ans. 0.80

Sol. Moles of $Cl_2 = \frac{710}{71} = 10$ mole
 $2Cl^- \longrightarrow Cl_2 + 2e^-$
 Means 20 F charge passed in this process, means 20 mole of NaOH would be formed.

$$\begin{aligned}\text{NaOH weight} &= 20 \times 40 \\ &= 800 \text{ gram} = 0.80 \text{ kg}\end{aligned}$$

36. Find the value of y_____

Ans. 2.24

Sol. $2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-$
 $20\text{F} \equiv 10 \text{ mole H}_2$
 Volume of H_2 at S.T.P. = 22.4×10
 $= 2.24 \times 10^2 \text{ L}$
 $y = 2.24$

Question Stem for Question Nos. 37 and 38

Question Stem

'x' gram of non-volatile solute urea needs to dissolve in 100 gram of water, due to this vapour pressure of water decreased by 25%. Finally molality of solution becomes y molal.

37. Find x_____

Ans. 111.11

Sol.
$$\frac{100 - 75}{100} = \frac{\left(\frac{x}{60}\right)}{\frac{x}{60} + \left(\frac{100}{18}\right)}$$

 $x = 111.11 \text{ gram}$

38. Find y_____

Ans. 18.52

Sol.
$$\begin{aligned}\text{Molality} &= \left(\frac{P^\circ - P_s}{P_s}\right) \left(\frac{1000}{\text{M.W. of solvent}}\right) \\ &= \left(\frac{100 - 75}{75}\right) \left(\frac{1000}{18}\right) \\ &= \frac{1000}{54} = 18.52 \text{ molal} \\ y &= 18.52\end{aligned}$$

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Let α, β are the roots of equation $x^2 - a_1x + 1 = 0$ and consider the sequence of numbers $a_r, r \geq 0$ with $a_0 = 1$ and $a_{r+1}^2 = 1 + a_r \cdot a_{r+2}$. Then which of the following is true?

(A) $a_r + a_{r+2} = a_1 \cdot a_{r+1}$

(B) $a_r + a_{r+2} = 2a_{r+1}$

(C) $a_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}$

(D) $a_n = \frac{\alpha^{n+1} + \beta^{n+1}}{\alpha + \beta}$

Ans. A, C

Sol. We have $a_r \cdot a_{r+2} = a_{r+1}^2 - 1 \forall r \geq 0$ and $a_0 = 1$

$$\text{So, } a_r + a_{r+2} = a_r + \frac{a_{r+1}^2 - 1}{a_r} = \frac{a_r^2 + a_{r+1}^2 - 1}{a_r} = \frac{a_r^2 - 1 + a_{r+1}^2}{a_r}$$

$$\Rightarrow \frac{a_{r-1} \cdot a_{r+1} + a_{r+1}^2}{a_r} = \frac{(a_{r-1} + a_{r+1}) \cdot a_{r+1}}{a_r} \dots = \frac{(a_0 + a_2) \cdot a_{r+1}}{a_1} = \frac{(1 + a_1^2 - 1) \cdot a_{r+1}}{a_1} = a_1 \cdot a_{r+1}$$

$$\Rightarrow a_r + a_{r+2} = a_1 \cdot a_{r+1} \text{ again } \alpha + \beta = a_1, \alpha\beta = 1$$

$$\text{So, } a_r + a_{r+2} = (\alpha + \beta) \cdot a_{r+1}; a_{r+2} - \beta a_{r+1} = \alpha a_{r+1} - a_r = \alpha a_{r+1} - \alpha\beta a_r$$

$$a_{r+2} - \beta a_{r+1} = \alpha(a_{r+1} - \beta a_r) \Rightarrow a_n - \beta a_{n-1} = \alpha(a_{n-1} - \beta a_{n-2})$$

$$a_n - \beta a_{n-1} = \alpha(a_{n-1} - \beta a_{n-2}) = \alpha^2(a_{n-2} - \beta a_{n-3}) = \alpha^{n-1}(a_1 - \beta a_0) = \alpha^{n-1}(a_1 - \beta)$$

$$a_n - \beta a_{n-1}; a_n - \beta a_{n-1} = \alpha^n; a_{n-1} - \beta a_{n-2} = \alpha^{n-1}; \beta(a_{n-1} - \beta a_{n-2}) = \beta \cdot \alpha^{n-1}$$

$\vdots$

$$\text{Add } \frac{\beta^{n-1}(a_1 - \beta a_0) = \beta^{n-1} \cdot \alpha}{a_n - \beta^n = \alpha^n + \beta \alpha^{n-1} + \beta^2 \alpha^{n-2} + \dots + \alpha \cdot \beta^{n-1}}$$

$$\Rightarrow a_n = \frac{\alpha^n \cdot \left(1 - \frac{\beta^{n+1}}{\alpha}\right)}{1 - \frac{\beta}{\alpha}} \Rightarrow a_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}$$

40. Let $\Delta(r) = \begin{vmatrix} \frac{1}{(r+2)^2} & \frac{1}{(r+2)} & \frac{-(r+1)}{(r+2)^2} \\ \frac{1}{(r+3)^2} & \frac{1}{r+1} & \frac{1}{(r+3)^2} - \frac{1}{(r+1)} \\ -2 & -1 & 0 \end{vmatrix}$ which of the following statements is(are)

CORRECT?

(A) $\sum_{r=1}^7 \Delta(r) = \frac{49}{900}$

(B) $\sum_{r=1}^{\infty} \Delta(r) = \frac{1}{18}$

(C) $\sum_{n=1}^7 \Delta(r) = \frac{49}{450}$

(D) $\sum_{r=1}^{\infty} \Delta(r) = \frac{1}{9}$

Ans. A, B

Sol. $C_3 \rightarrow C_3 + C_2 - C_1$

$$\Delta(r) = \begin{vmatrix} \frac{1}{(r+2)^2} & \frac{1}{(r+2)} & 0 \\ \frac{1}{(r+3)^2} & \frac{1}{r+1} & 0 \\ -2 & -1 & 1 \end{vmatrix}; \Delta(r) = \frac{1}{(r+1)(r+2)^2} - \frac{1}{(r+2)(r+3)^2}$$

$$\Delta(r) = \left(\frac{1}{2 \cdot 3^2} - \frac{1}{3 \cdot 4^2} \right) + \left(\frac{1}{3 \cdot 4^2} - \frac{1}{4 \cdot 5^2} \right) \dots \left(\frac{1}{(n+1)(n+2)^2} - \frac{1}{(n+2)(n+3)^2} \right)$$

$$= \frac{1}{2 \cdot 3^2} - \frac{1}{(n+2)(n+3)^2}; n = 7 = \frac{1}{2 \cdot 9} - \frac{1}{9 \cdot 100} = \frac{49}{900} = \frac{1}{9}$$

41. Let $t(r) = \frac{{}^{n+r}C_r (2r-1)}{{}^nC_r \cdot (n+r)}$, then which of the following are true? (Where $v(r) = \frac{{}^{n+r}C_r}{{}^nC_r}$)

(A) $\sum t_r = {}^{2n}C_n - 1$

(B) $t(r) = v(r) - v(r-1)$

(C) $\sum t_r = {}^{2n}C_n$

(D) $\sum t_r = \frac{2^n (1 \cdot 3 \cdot 5 \dots (2n-1))}{n!}$

Ans. A, B

Sol. $t_r = \frac{{}^{n+r}C_r \cdot (2r-1)}{{}^nC_r \cdot (n+r)} = \frac{{}^{n+r}C_r}{n} \left(1 - \frac{n-r+1}{n+r}\right) = \frac{{}^{n+r}C_r}{n} - \frac{{}^{n+r}C_r}{n} \left(\frac{n-r+1}{n+r}\right)$

Now, $\frac{{}^{n+r}C_r}{n} \left(\frac{n-r+1}{n+r}\right) = \frac{(n-r+1)(n+r)!r!(n-r)!}{r!n!(n+r)n!} = \frac{(n-r+1)!(n+r-1)!}{n!n!} = \frac{{}^{n+r-1}C_{r-1}}{n}$

$t_r = \frac{{}^{n+r}C_r}{n} - \frac{{}^{n+r-1}C_{r-1}}{n} \Rightarrow \sum t_r = {}^{2n}C_n - 1$

42. Let $x + y + 3z - 4 = 0$, $x + 2y + bz - 5 = 0$ and $2x + ay + 6z - 8 = 0$ be three planes, then

- (A) three planes intersect at a point if $a \neq 2$, $b \neq 3$
- (B) three planes do not have any common point of intersection if $a \neq 2$, $b = 3$
- (C) three planes have infinite points common among them if $a = 2$, $b \in \mathbb{R}$
- (D) three planes never intersect at exactly two distinct points

Ans. A, B, C, D

Sol. Unique solution if $a \neq 2$ and $b \neq 3$
 $\Rightarrow$ No solution if $a \neq 2$ and $b = 3$
 $\Rightarrow$ Infinite solution if $a = 2$, $b \in \mathbb{R}$

43. If $(1+x+x^2)^6 = A_0x^6 + (A_1x^2 + B_1)x^5 + (A_2x^4 + B_2)x^4 + \dots + (A_6x^{12} + B_6)$, then

- (A) $A_0 = 141$
- (B) $A_5 = 6$
- (C) $\sum_{i=1}^6 (A_i + B_i) = 588$
- (D) $\sum_{i=1}^6 (A_i + B_i) = 3^6$

Ans. A, B, C

Sol. $(1+x+x^2)^6 = x^6 \left(A_0 + \left(A_1x + \frac{B_1}{x} \right) + \left(A_2x^2 + \frac{B_2}{x^2} \right) + \dots + \left(A_6x^6 + \frac{B_6}{x^6} \right) \right)$

$\therefore \left(x + \frac{1}{x} + 1 \right)^6 = \sum_{r=1}^6 {}^6C_r \cdot \left(x + \frac{1}{x} \right)^r$

(A) $A_0 = 1 + {}^6C_2 \cdot {}^2C_1 + {}^6C_4 \cdot {}^4C_2 + {}^6C_5 \cdot {}^6C_3 = 141$

(B) $A_5 = {}^6C_2$ coefficient of x^5 in $\left(x + \frac{1}{x} \right)^5 = {}^6C_5 \cdot {}^5C_0 = 6$

$\Rightarrow \sum_{i=1}^6 (A_i + B_i) = 3^6 - 141$

44. The necessary and sufficient conditions that the equation $\begin{vmatrix} 0 & a_1 - x & a_2 - x \\ -a_1 - x & 0 & a_3 - x \\ -a_2 - x & -a_3 - x & 0 \end{vmatrix} = 0$, where $a_i \neq 0$ shall have a multiple roots

- (A) $a_1a_2 + a_2a_3 + a_3a_1 = 0$
 (B) $a_1a_2 + a_2a_3 - a_1a_3 = 0$
 (C) $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} = 0$
 (D) $\frac{1}{a_1} + \frac{1}{a_3} = \frac{1}{a_2}$

Ans. B, D

Sol. Given determinant is $-2x^3 + 2(a_1a_2 + a_2a_3 - a_1a_3)x = 0$
 So, condition is $a_1a_2 + a_2a_3 - a_1a_3 = 0$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

For a polynomial $a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \dots + a_0 = 0$. If $x_1, x_2, x_3, \dots, x_n$ are n roots, then

$$\begin{aligned} \sum x_i &= \frac{(-1)^1 \cdot a_{n-1}}{a_n} \\ \sum x_i \cdot x_j &= \frac{(-1)^2 \cdot a_{n-2}}{a_n} \\ &\vdots \\ x_1 \cdot x_2 \cdot \dots \cdot x_n &= \frac{(-1)^n \cdot a_0}{a_n} \end{aligned}$$

45. Let $P(x)$ be a quadratic polynomial with complex coefficient whose leading coefficient 1. Suppose the equation $P(P(x)) = 0$ has 4 distinct solution, $x = 3, 4, \alpha, \beta$, then the sum of all possible values of $(\alpha + \beta)^2$ is
- (A) 49
 (B) 36
 (C) 85
 (D) 70

Ans. C

Sol. Let $P(x) = x^2 + ax + b$ since $P(P(x)) = 0$ has roots 3, 4

$$\Rightarrow P(P(3)) = 0 = P(P(4))$$

Case-I: $P(3) \neq P(4)$ and $P(3)$ and $P(4)$ will be roots of $P(x) = 0$

$$\text{So, } P(3) + P(4) = -a \text{ and } P(3)P(4) = b \Rightarrow a = -\frac{1}{2} \text{ and } b = -\frac{21}{2}$$

$$\text{So, } (\alpha + \beta)^2 = 36$$

Case-II: $P(3) = P(4) \Rightarrow \alpha + \beta = -7$

$$\text{So, } (\alpha + \beta)^2 = 49$$

So, required sum is 85

46. If a, b, c be the roots of the equation $x^3 + px + q = 0$, then $\frac{6(a^5 + b^5 + c^5)}{(a^3 + b^3 + c^3)(a^2 + b^2 + c^2)}$ is

(A) 5

(B) 3

(C) 2

(D) 1

Ans. A

Sol. (Q.45-46)

Since, a, b, c are roots of the equation $x^3 + px + q = 0$ (1)

$$\Rightarrow a^3 + pa + q = 0 \text{ (2)}$$

$$b^3 + pb + q = 0 \text{ (3)}$$

$$c^3 + pc + q = 0 \text{ (4)}$$

From equation (2), (3) (4), we get $\sum a^3 + p\sum a + 3q = 0$

$$\Rightarrow \sum a^3 = -32 \because \sum a = 0 \text{ also } \sum a^2 = (\sum a)^2 - 2\sum a \cdot b = -2p$$

Now, multiply x^2 in equation (2), we get $x^5 + px^3 + qx^2 = 0$ also has roots a, b, c

$$\sum a^2 + p\sum a^3 + 2\sum a^2 = 0 ; \sum a^5 + p(-32) + q(-2p) = 0$$

$$\sum a^5 = 5pq ; \frac{1}{5}\sum a^5 = \frac{\sum a^3}{(-3)} \cdot \frac{\sum a^2}{(-2)} \Rightarrow \frac{6\sum a^5}{\sum a^3 \sum a^2} = 5$$

Paragraph for Question Nos. 47 and 48

If $\vec{a}$ and $\vec{b}$ are two vectors, then $\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta$ and $\vec{a} \times \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \sin \theta \hat{n}$ where θ is angle between two vectors, then

47. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $|\vec{c}| = 5$ satisfying $[\vec{a}, \vec{b}, \vec{c}] = 30$, then $(2\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} \times \vec{c} \times (\vec{a} - \vec{c}) + \vec{b})$ is equal to

(A) 308

(B) 309

(C) 310

(D) 311

Ans. B

Sol. $|\vec{a}||\vec{b}||\vec{c}|\cos\phi\cdot\sin\theta = 30$; $\sin\theta\cos\phi = 1 \Rightarrow \theta = \frac{\pi}{2}, \phi = 0$

$\Rightarrow \vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular

$$\text{So, } (2\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} \times \vec{c} \times (\vec{a} - \vec{c}) + \vec{b}) \Rightarrow 2|\vec{a}|^2|\vec{c}|^2 + |\vec{b}|^2 + |\vec{a}|^2|\vec{c}|^2 = 309$$

48. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors satisfying $\vec{a} = \vec{b} \times \vec{c} + 2\vec{b}$ where $|\vec{b}| = |\vec{c}| = 2$ and $|\vec{a}| \leq 4$, then the sum of possible value of $|2\vec{a} + \vec{b} + \vec{c}|$ is

- (A) 8
- (B) 12
- (C) 20
- (D) 32

Ans. C

Sol. $\vec{a} = \vec{b} \times \vec{c} + 2\vec{b}$ taking dot product with $\vec{b}$

$$\vec{a} \cdot \vec{b} = 2|\vec{b}|^2 \Rightarrow |\vec{a}| \cdot |\vec{b}| \cos\theta = 2|\vec{b}|^2 \Rightarrow \cos\theta = \frac{4}{|\vec{a}|} \Rightarrow |\vec{a}| = 4 \Rightarrow \theta = 0$$

$$\Rightarrow \vec{a} = 2\vec{b} \because \vec{b} \times \vec{c} = 0 \Rightarrow \vec{b} = \vec{c} \text{ or } \vec{b} = -\vec{c}$$

$$\text{So, } |2\vec{a} + \vec{b} + \vec{c}| \Rightarrow |\vec{a}| = 12 \text{ and } 2|\vec{a}| = 8$$

$$\therefore 12 + 8 = 20$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. For any increasing sequence of positive integers $a_1, a_2, a_3, \dots$ has the property that for every positive integer k , the subsequence $a_{2k-1}, a_{2k}, a_{2k+1}$ is in geometric sequence and the subsequence $a_{2k}, a_{2k+1}, a_{2k+2}$ is in Arithmetic sequence, if $a_{13} = 2016$, then a_1 is _____

Ans. 504

Sol. Let $a_1 = a$ and $a_2 = ka$

Sequence becomes $a, ka, k^2a, k(2k-1)a, (2k-1)^2a, (2k-1)(3k-2)a, (3k-2)^2a$

$$\text{So, } a_{2k+1} = (nk - (n-1))^2 a$$

$$\Rightarrow a_{13} = (6k-5)^2 a = 2016$$

$$\Rightarrow a = \frac{2016}{(6k-5)^2} = \frac{2016}{2^2} = 504$$

50. The nine delegates in a conference included 2 from Pakistan, 3 from China and 4 from United States. During the opening session, three of the delegates fall asleep. Assuming that the three sleepers were determined randomly. The probability that exactly two of the sleepers are from the same country in $\frac{a}{b}$, where a and b are relative prime, then $a + b$ is _____

Ans. 139

Sol. Required probability = $\frac{55}{84}$

51. Six cards numbered 1 to 6 are to be lined up in a row. Then the number of arrangements of these six cards where one of the cards can be removed leaving the remaining five cards in either ascending or descending order

Ans. 52

Sol. Ascending can be reversed for descending so taking one order
i.e., if we choose any of the numbers 1 to 6 these are five other place to put them
So, we get $6 \times 5 = 30$. But some case repeated
So, repeated case are only 5
Total = $30 - 5 = 25$ add one more for original case = $25 + 1 = 26$
 $\Rightarrow 26 \times 2 = 52$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = (a_{ij}(n))$, then

52. $\lim_{n \rightarrow \infty} \frac{a_{12}(n)}{a_{22}(n)}$ is equal to _____

Ans. 2.41

53. $a_{12}(n)$ for $n = 2$ is equal to _____

Ans. 2.00

Sol. (Q.52-53)

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_I + \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}_B \right)^n$$

$$A = (I + B)^n = {}^nC_0 \cdot I + {}^nC_1 B + {}^nC_2 \cdot B^2 + {}^nC_3 \cdot B^3 + \dots + {}^nC_n B^n$$

$$\text{Since, } B^2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

$$B^{2k} = 2^k \cdot I \text{ and } B^{2k+1} = B^{2k} \cdot B = 2^k \cdot B$$

$$\Rightarrow A = \underbrace{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n}_{x} = ({}^nC_0 + {}^nC_2 \cdot 2 + {}^nC_4 \cdot 2^2 + \dots)I + \underbrace{[{}^nC_1 + {}^nC_3 \cdot 2 + {}^nC_5 \cdot 2^2 \dots]}_y B$$

$$\begin{aligned}
 &= \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & y \\ y & -y \end{bmatrix} = \begin{bmatrix} x+y & y \\ y & x-y \end{bmatrix} \\
 &\frac{a_{12}}{a_{22}} = \frac{y}{x-y} = \frac{({}^nC_1 + {}^nC_3 \cdot 2 + {}^nC_5 \cdot 2^2 + \dots)}{({}^nC_0 + {}^nC_2 \cdot 2 + {}^nC_4 \cdot 2^2 + \dots) - ({}^nC_1 + {}^nC_3 + \dots)} \\
 &\quad \frac{1}{\sqrt{2}} \frac{((1+\sqrt{2})^n - (1-\sqrt{2})^n)}{2} \\
 &\lim_{n \rightarrow \infty} \frac{2}{\frac{(\sqrt{2}-1)(1+\sqrt{2})^n - (\sqrt{2}+1)(1-\sqrt{2})^n}{2\sqrt{2}}} = 1 + \sqrt{2}
 \end{aligned}$$

$$a_{11} \text{ is } {}^nC_0 + {}^nC_2 \cdot 2 + {}^nC_4 \cdot 2^2 + \dots; {}^nC_1 + {}^nC_3 \cdot 2 + {}^nC_5 \cdot 2^2 + \dots$$

$$\frac{1}{\sqrt{2}} ({}^nC_1 \cdot \sqrt{2} + {}^nC_3 (\sqrt{2})^3 + {}^nC_5 (\sqrt{2})^5 + \dots) = \frac{1}{\sqrt{2}} \frac{((1+\sqrt{2})^n - (1-\sqrt{2})^n)}{2}$$

Question Stem for Question Nos. 54 and 55

Question Stem

If $y = \log_a x$, $x > 0$, $a > 0$, $a \neq 1$

54. For each ordered pairs of real number x, y satisfying $\log_2(2x+y) = \log_4(x^2+xy+7y^2)$ there is a real number k such that $\log_3(3x+y) = \log_9(3x^2+4xy+ky^2)$, then the product of all possible values of k is

Ans. 189.00

Sol. Since $(2x+y)^2 = x^2+xy+7y^2 \Rightarrow x=y$ or $x=-2y$ and for that $k=9, 21$
Product = 189

55. There is a unique positive real x such that the three numbers $\log_8 2x, \log_4 x, \log_2 x$ in that order form a geometric progression with positive common ratio, if $x = \frac{m}{n}$, where m, n relative prime positive integer, then $m+n$ is

Ans. 17.00

Sol. These form a Geometric Progression, $\frac{\log_2 x}{\log_4 x}$ is common ratio = 2

$$\text{Now, } \frac{\log_4 x}{\log_8 2x} = 2 \Rightarrow \log_4 x = 2 \log_8 2x$$

$$\frac{1}{2} \log_2 x = 2 \log_8 2 + 2 \log_8 x; \frac{1}{2} \log_2 x = \frac{2}{3} + \frac{2}{3} \log_2 x$$

$$\left(\frac{1}{2} - \frac{2}{3}\right) \log_2 x = \frac{2}{3}; -\frac{1}{6} \log_2 x = \frac{2}{3}; \log_2 x = -\frac{2 \times 6}{3} = -4$$

$$\therefore x = 2^{-4} = \frac{1}{16}$$

Question Stem for Question Nos. 56 and 57

Question Stem

Let z be a complex number and $z = a + ib$, $a, b \in \mathbb{R}$ (where $i = \sqrt{-1}$)

56. Let N be the number of complex numbers z with the properties that $|z| = 1$ and $z^{6!} - z^{5!}$ is a real number. Find remainder when N is divided by 1000

Ans. 440.00

Sol. Since $|z| = 1$, let $z = \cos \theta + i \sin \theta$ so, $z^{720} - z^{120}$ is real

$$\Rightarrow \sin 720\theta - \sin 120\theta = 0 \Rightarrow 2 \cos \frac{840\theta}{2} \cdot \sin \frac{600\theta}{2} = 0$$

$$\Rightarrow \cos \frac{840}{2}\theta = 0 \text{ or } \sin \frac{600}{2}\theta = 0 \Rightarrow \theta = 840 \text{ or } \theta = 600$$

Total = 1440 solution so option is 440

57. For all complex number z with $|z| = 1$, the maximum of $|1 - z| + |1 + z^2|$ is

Ans. 4.00

Sol. Let $z = \cos \theta + i \sin \theta$; $|1 - z| = 2 \left| \sin \frac{\theta}{2} \right|$; $|1 + z^2| = 2 \left| 1 - 2 \sin \frac{\theta}{2} \right|$

$$\Rightarrow |1 - z| + |1 + z^2| \leq 4$$